REAL-TIME DRONE DETECTION SYSTEM

AI-Based Object Detection & Edge Deployment

Technical Project Documentation

Model: YOLO26n
Framework: PyTorch & Ultralytics
Deployment Device: MaixCAM

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1. Introduction

The Real-Time Drone Detection System is an AI-based computer vision project designed to detect drones through a camera feed.

The project uses the YOLO26n object detection model, trained using Python and the Ultralytics framework. The trained model is exported into ONNX format and converted through MaixHub for deployment on the MaixCAM edge AI device.

The main goal is to develop a lightweight detection system capable of identifying drones in real-time using edge hardware.

2. Project Objectives

- Develop an AI-based drone detection model.
- Train the model using a custom drone detection dataset.
- Detect drones from camera input in real-time.
- Export the trained model into ONNX format.
- Deploy and test the converted model on MaixCAM.
- Explore edge AI deployment for real-world applications.

3. Technologies & Tools

Technology	Purpose
Python	Main programming language
PyTorch	Deep learning framework
Ultralytics	YOLO model training and inference

YOLO26n	Drone object detection
Roboflow	Dataset preparation and annotation
OpenCV	Computer vision and image processing
VS Code	Development environment
ONNX	Model export format
MaixHub	Model conversion
MaixCAM	Edge AI deployment and testing

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4. Model Information

The project uses **YOLO26n (Nano)**, a lightweight object detection model from the Ultralytics YOLO26 family.

The Nano variant was selected for its compact architecture and suitability for applications where computational resources are limited.

The model was trained to identify drones from images and visual input.

5. Training Configuration

The following configuration was used during model training:

Parameter	**Value**
Model	YOLO26n
Image Size	1280 × 1280
Epochs	85
Batch Size	8
Workers	2
Patience	21
Framework	PyTorch

Library	Ultralytics

6. Training Process

The model training process involved the following steps:

1. Preparing and organizing the drone detection dataset.
2. Preparing image annotations for object detection.
3. Configuring the YOLO26n model.
4. Training the model using the Ultralytics framework.
5. Saving the best-performing model weights.
6. Exporting the trained model into ONNX format.

The training process produces the best model checkpoint, commonly saved as best.pt.

The exported model used in the deployment workflow is named best.onnx.

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7. Model Export & Deployment

After training, the model was exported from PyTorch format into ONNX format.

ONNX provides a standard model format that helps transfer trained models between different software frameworks and deployment platforms.

The deployment workflow is:

Training → best.pt → ONNX Export → best.onnx → MaixHub Conversion → MaixCAM

MaixHub was used to convert the ONNX model into a format suitable for the target MaixCAM device.

The converted model was then used for testing on MaixCAM.

8. Real-Time Testing

The trained and converted model was tested on the MaixCAM device.

The testing focused on:

- Live camera input.
- Real-time drone detection.

- Object localization using bounding boxes.
- Displaying detection results.
- Running AI inference directly on edge hardware.

The project demonstrates the deployment of a custom-trained drone detection model on an edge AI device.

9. Future Improvements

Possible future improvements include:

- Improving detection accuracy for small and distant drones.
- Testing in different lighting and weather conditions.
- Optimizing inference speed on edge hardware.
- Expanding and improving the training dataset.
- Evaluating model performance using precision, recall and mAP.
- Testing detection performance in different real-world environments.

10. Conclusion

This project demonstrates the development of an AI-based real-time drone detection system using YOLO26n.

By combining Python, PyTorch, Ultralytics, Roboflow, OpenCV and ONNX, the model was trained and prepared for edge deployment.

The use of MaixHub and MaixCAM provided a practical way to test the trained model on dedicated edge AI hardware.

The project provides a foundation for further development of lightweight, real-time drone detection applications.

11. Technical References

1. Ultralytics YOLO26: https://docs.ultralytics.com/models/yolo26/
2. Ultralytics Model Export: https://docs.ultralytics.com/modes/export/
3. Roboflow Documentation: https://docs.roboflow.com/
4. MaixCAM Documentation: https://wiki.sipeed.com/maixpy/